

Tesla, Inc. (NASDAQ: TSLA)

From hardware OEM to physical-AI and distributed-energy platform: a 10-year view

Rating: BUY (long-term) Price / Mkt cap: ~\$220 / ~\$700B **2035 target: \$520 base (12.8% IRR)** Bull \$950 / Bear \$180

EXECUTIVE SUMMARY

Most of the market still values Tesla as a car company that happens to have an AI narrative attached. My view is the reverse: by 2035 Tesla is an AI and energy-infrastructure business that still sells cars. The vehicle franchise is mature and margin-compressed, but it is also the thing that funds and trains everything else, and it is the reason Tesla can attempt what pure software firms cannot.

Three transitions drive the 10-year case: (1) Full Self-Driving moves from a one-off option to a recurring subscription and a robotaxi network, lifting group operating margin from ~8% toward 22%; (2) Megapack turns grid instability into a contracted, utility-grade cash flow stream; (3) Optimus, the longest-dated and least certain leg, gives real upside optionality. A net-cash balance sheet of \$35B+ means the bear case is a poor decade, not a solvency event.

Exhibit 1. Historical financial snapshot (Tesla 10-K filings; FY25E = analyst estimate)

\$ millions	FY23	FY24	FY25E	Read-through
Total revenue	96,773	97,690	104,500	EV plateau; storage +40% YoY
Auto GM ex-credits	18.2%	17.9%	18.1%	HW margins bottoming
Energy storage rev.	6,035	10,071	14,800	2nd engine; GM to ~29%
Free cash flow	4,358	3,580	6,200	Funds \$10B+ AI capex
Cash & ST inv.	29,100	36,500	41,200	Net cash, no solvency risk

1. BUSINESS MODEL AND WHERE THE MONEY GOES

Revenue mix today vs. 2035. Roughly 78% of 2025 revenue is vehicle sales and 22% is energy plus services. The 2035 base case assumes EV hardware falls to ~35% of a much larger pie, with mobility services (FSD subscriptions, robotaxi rides, OEM licensing) at ~30%, Megapack and virtual-power-plant software at ~20%, and Optimus at ~15% (Exhibit 2). The point is not the exact split but the direction: the incremental dollar is increasingly software or contracted infrastructure, not sheet metal.

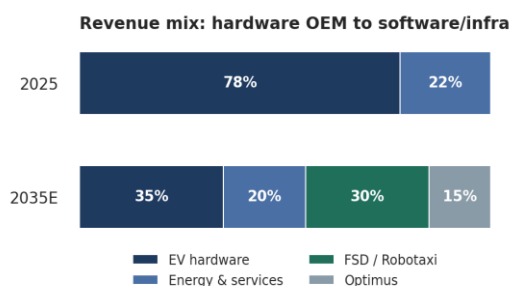


Exhibit 2. Revenue mix shift, 2025 vs. 2035E (own estimates)

Key revenue and cost drivers. On the cost side, the 'unboxed' manufacturing process for the next-generation platform targets ~30% lower cost per vehicle and roughly half the capex per unit of capacity. On the revenue side, FSD is the swing factor: at \$99/month, a 65% attach rate on a fleet that will exceed 10 million vehicles is a multi-billion-dollar, ~80%-gross-margin annuity that did not exist three years ago.

2. CASH FLOW DEFENSIVENESS

Tesla's cash flow is more defensive than a cyclical automaker's for two reasons. First, the balance sheet: \$36.5B of cash and short-term investments at FY24 against minimal net debt, with FCF positive every year since 2019 despite \$10B+ of annual capex and heavy AI compute spend. Second, the mix is migrating toward recurring revenue: FSD subscriptions, and Megapack contracts that typically bundle 10-to-15-year service and software agreements with utilities, which are not tied to the consumer car cycle. Even if vehicle demand disappoints, the company can fund its AI roadmap internally.

3. INDUSTRY POSITION, MOAT AND ESG

Moat. Tesla is the only automaker that owns the full stack: vehicle, in-house inference chip, training compute, and billions of real-world miles of driving data collected from customer cars. Competitors buy chips from NVIDIA and driving data from simulation; Tesla's fleet generates both scale and edge-case data at near-zero marginal cost. In energy, Megafactories in Lathrop and Shanghai give it the largest dedicated stationary-storage capacity in the West.

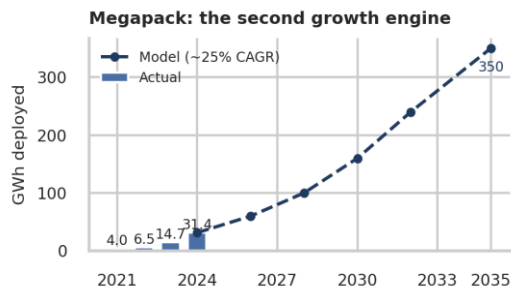
ESG. Environmental benefit is straightforward: every GWh of Megapack deployed displaces gas peaker capacity, and the EV fleet is the largest single contributor to transport decarbonization in North America. The governance picture is the honest weak spot. Board independence is thin, and the CEO's political visibility has produced measurable brand damage in Western Europe (registrations in France, Germany

and Spain fell sharply through 2025). I treat this as a real, quantifiable headwind rather than something to explain away, and it is one reason my base-case EV volume is conservative.

4. INVESTMENT THESIS: THREE PILLARS

Pillar 1 - FSD and robotaxi re-rate the multiple. The shift to end-to-end neural networks (v12 and later) replaced hand-written rules with a system that improves with data, and the Austin robotaxi pilot launched in mid-2025 marks the first paid autonomous rides. As regulatory approvals widen, each incremental subscriber or ride carries almost no marginal cost. This is what takes operating margin from ~8% to 22-25% and, just as important, changes how investors frame the stock: from a 15x-EBITDA manufacturer toward a 25x+ platform.

Pillar 2 - Megapack monetizes grid instability. Solar and wind are now the cheapest new generation almost everywhere, but they are intermittent, and utilities pay for firmness. Deployments rose from 14.7 GWh in 2023 to 31.4 GWh in 2024; with Shanghai ramping, I model ~350 GWh by 2035 at a ~25% CAGR (Exhibit 3). At a ~29% segment gross margin, energy should contribute more than a quarter of operating income within five years, and it is insulated from consumer sentiment about the brand.



7. KEY RISKS AND HOW I THINK ABOUT THEM

- Regulatory or safety delay to FSD.** The core risk. Mitigant: Megapack growth and the EV franchise support a valuation near today's price even if robotaxi commercialization slips 3-5 years.
- Brand backlash and key-person risk.** Real in Europe and hard to reverse. Mitigant: lower-priced Standard trims broaden the funnel, and B2B energy and fleet contracts are largely indifferent to consumer politics.
- China price war and margin compression.** BYD and peers keep pressure on ASPs. Mitigant: a 95%+ localized Shanghai supply chain and the unboxed process protect unit economics better than any Western peer.

Conclusion. Tesla is not a cheap stock on today's earnings and it will not be a smooth decade. But the combination of a fortress balance sheet, a second growth engine that is already compounding, and a credible path to software-like margins gives a return profile I am willing to own for ten years. BUY, base-case target \$520 by 2035.

Exhibit 3. Energy storage deployments, GWh (Tesla reports; 2026+ own model)

Pillar 3 - Optimus is the option, not the thesis. Tesla already builds motors, batteries, and inference chips at scale, which is why a humanoid robot is a plausible adjacency rather than a distraction. The path runs through its own factories first. I assume 500k units per year by 2035, worth ~15% of revenue; the number could be zero or several times higher, and I deliberately give it little weight in the base case.

5. KPI TRAJECTORY TO 2035

- FSD fleet penetration:** ~12% (2025) to 65%+ (2035E), the single most important driver of margin.
- Storage deployments:** 31.4 GWh (2024) to 350+ GWh (2035E), ~25% CAGR.
- Operating margin:** ~8% to 21.5% (base), as software and contracted infrastructure outgrow hardware.
- ROIC:** ~10% to ~22%, as the capex-heavy build-out is completed and recurring revenue matures.

6. VALUATION

I use a 10-year DCF (9.0% WACC, 3.0% terminal growth) cross-checked against a 2035 sum-of-the-parts and probability-weighted scenarios (Exhibit 4). The base case assumes \$380B revenue and a 21.5% operating margin at a 25x exit multiple, giving \$520 per share and a 12.8% annual return. What makes the risk/reward attractive is the asymmetry: the bear case (robotaxi never scales, EV margins stay compressed) roughly returns the current price because the balance sheet and the energy business hold the floor, while the bull case is 3-4x. A probability-weighted target of ~\$538 implies ~13% annual return, which is adequate compensation for the execution risk.

Exhibit 4. 2035E scenario analysis (DCF/SOTP; own estimates)

2035E scenario	Bear (20%)	Base (60%)	Bull (20%)
Revenue	\$180B	\$380B	\$650B
Operating margin	11.0%	21.5%	28.0%
Exit EV/EBITDA	15.0x	25.0x	38.0x
Target price	\$180	\$520	\$950
10-yr IRR	-2.0%	+12.8%	+19.5%

Probability-weighted target: 0.2x\$180 + 0.6x\$520 + 0.2x\$950 = \$538. Sources: Tesla 10-K/10-Q filings and quarterly deployment reports; all forward figures are the author's estimates.